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Chemistry
Class 10

Sindh Textbook Board

(English Medium)

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Chemical Equilibrium

Unit 2
Acids, Bases and Salts

Unit 3
Organic Chemistry

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Biochemistry

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Chemical Equilibrium

UNIT 1

Reversible Reactions and Dynamic Equilibrium

LESSON 1

Many physical and chemical changes take place around us. Some changes can be reversed and some cannot. In this lesson we will understand reversible reactions and dynamic equilibrium.

1. Reversible Reaction

A reaction which never goes to completion and proceeds in both the forward and the reverse directions is called a reversible reaction. It is represented by a double arrow <=>.

- Melting of ice to form water and freezing of water to form ice is a reversible change.
- Condensation, evaporation, freezing and melting are reversible changes.
- Example: N2 + O2 <=> 2NO — the forward reaction forms NO from N2 + O2 and the reverse reaction forms N2 + O2 again from NO.

2. Irreversible Reaction

A reaction which proceeds in only one direction and goes to completion is called an irreversible reaction. It is represented by a single arrow ->. Example: burning of fuel (combustion) and rusting of iron.

Reversible Reaction	Irreversible Reaction
Proceeds in both directions	Proceeds in only one direction
Never goes to completion	Goes to completion
Represented by double arrow <=>	Represented by single arrow ->
Can establish equilibrium	Does not establish equilibrium

3. Chemical Equilibrium

Chemical equilibrium is the state in which, in a reversible reaction in a closed container, the rate of the forward reaction becomes equal to the rate of the reverse reaction and the concentrations of the reactants and products stay constant. Example: H2 + I2 <=> 2HI.

4. Why Is Equilibrium Dynamic?

- At equilibrium the two reactions do not stop.
- The forward and reverse reactions keep going at equal rates.
- No change is seen from outside, but internally the reaction continues.
- Equilibrium can be reached from either side.
- Equilibrium can be disturbed by external stress.

5. Equilibrium in Our Body

Equilibrium is also present in our body. Haemoglobin is a large molecule which carries oxygen to the whole body. Haemoglobin has to both pick up oxygen and release it, and this work takes place at different places in the body through changes in the chemical equilibrium of this reaction.

Law of Mass Action and Expression of Equilibrium Constant

LESSON 2

1. Active Mass

Active mass is the molar concentration of a reacting substance. Its unit is mol dm^-3 and it is written in square brackets, for example [H2].

2. Law of Mass Action

At any instant, the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants at that instant.

3. Deriving the Expression for Kc

For the general reaction: aA + bB <=> cC + dD

- Rate of forward reaction = kf [A]^a [B]^b
- Rate of reverse reaction = kr [C]^c [D]^d
- At equilibrium the two rates are equal: kf [A]^a [B]^b = kr [C]^c [D]^d
- Therefore Kc = kf / kr = [C]^c [D]^d / [A]^a [B]^b

That is, in Kc the concentrations of the products raised to the power of their coefficients are written on top and the concentrations of the reactants raised to the power of their coefficients are written below.

4. Examples

- For S + O2 <=> SO2: Kc = [SO2] / [S] [O2]
- For N2 + 3H2 <=> 2NH3: Kc = [NH3]^2 / [N2] [H2]^3
- For 4NH3 + 5O2 <=> 4NO + 6H2O: Kc = [NO]^4 [H2O]^6 / [NH3]^4 [O2]^5

5. Reaction Quotient Qc

Qc is the ratio of the molar concentrations of products and reactants raised to the power of the coefficients, but it is measured at any particular instant, not necessarily at equilibrium. Example: for H2 + I2 <=> 2HI, Kc = 48 (at 717 K). If [H2] = 0.2, [I2] = 0.2 and [HI] = 0.6 mol dm^-3, then Qc = [HI]^2 / [H2] [I2] = 0.36 / 0.04 = 9.

Equilibrium Constant and Its Units

LESSON 3

1. Characteristics of Kc

- Kc works only at equilibrium.
- It expresses the equilibrium concentrations of reactants and products in mol dm^-3.
- It is independent of the initial concentrations.
- It changes with temperature, and temperature is the single factor affecting its value.
- Kc tells both the direction and the extent of the reaction.

2. When Does Kc Have a Unit?

Whether Kc has a unit depends on whether the numbers of moles on the two sides are equal or not.

Condition	Example	Unit of Kc
Moles on both sides are equal	N2 + O2 <=> 2NO	No unit
Moles on both sides are not equal	N2 + 3H2 <=> 2NH3	mol^-2 dm^6
Two moles of product from one mole	N2O4 <=> 2NO2	mol dm^-3

3. Worked Example 1

For N2O4 <=> 2NO2, Kc = [NO2]^2 / [N2O4]. Unit: the top has mol dm^-3 squared and the bottom has mol dm^-3, so the unit of Kc is mol dm^-3.

4. Worked Example 2

For PCl5 <=> PCl3 + Cl2, at 500 K [PCl5] = 0.8 x 10^-3, [PCl3] = 1.2 x 10^-3 and [Cl2] = 1.2 x 10^-3 mol dm^-3. Kc = [PCl3] [Cl2] / [PCl5] = (1.2 x 10^-3 x 1.2 x 10^-3) / (0.8 x 10^-3) = 1.8 x 10^-3 mol dm^-3.

Importance of Equilibrium Constant

LESSON 4

1. What Does the Magnitude of Kc Tell Us?

Magnitude of Kc	Meaning
Small Kc	The reaction mixture mostly contains reactants
Moderate Kc	The mixture contains almost equal amounts of reactants and products
Large Kc	The reaction mixture mostly contains product

- If Kc is greater than 1, the reaction is in the forward direction.
- If Kc is less than 1, the reaction is in the reverse direction.
- Kc tells both the direction and the extent of the reaction.

2. Example: Two Systems

- For 2CO2 <=> 2CO + O2, Kc = 3.1 x 10^-18 mol dm^-3 (very small) — at equilibrium there will be mostly reactant CO2 and very little product.
- For 2O3 <=> 3O2, Kc = 5.9 x 10^55 mol dm^-3 (very large) — at equilibrium there will be mostly product O2 and very little reactant.

3. Predicting Direction by Comparing Qc and Kc

Condition	Result
Qc < Kc	The net reaction proceeds forward, more product is formed
Qc > Kc	The net reaction proceeds in reverse, more reactants are formed
Qc = Kc	The system is in equilibrium, no net change occurs

4. Worked Example

For 2HI <=> H2 + I2, Kc = 1 x 10^-4. If [HI] = 2 x 10^-5, [H2] = 1 x 10^-5 and [I2] = 1 x 10^-5 mol dm^-3, then Qc = [H2] [I2] / [HI]^2 = 1 x 10^-10 / 4 x 10^-10 = 0.25. Qc, that is 0.25, is greater than Kc, so the net reaction will proceed in the reverse direction and more HI will be formed.

Acids, Bases and Salts

UNIT 2

Concepts of Acids and Bases

LESSON 5

1. Properties of Acids and Bases

Acids	Bases
Sour in taste	Bitter in taste
Give H+ in aqueous solution	Give OH- in aqueous solution
Turn blue litmus paper red	Turn red litmus paper blue
Have a corrosive effect	Feel soapy

Note: Both acids and bases can have a corrosive effect on the skin.

2. Arrhenius Concept

According to Arrhenius, an acid is one that gives H+ in aqueous solution, for example HCl -> H+ + Cl-, and a base is one that gives OH- in aqueous solution, for example NaOH -> Na+ + OH-. This theory is only applicable to aqueous solutions, because without water these ions are not formed. For example, CO2 is not an Arrhenius acid.

3. Bronsted-Lowry Concept

According to this theory, an acid is a proton (H+) donor and a base is a proton acceptor. Example: In HCl + NH3 -> NH4+ + Cl-, HCl donates a proton so it is the acid, and NH3 accepts a proton so it is the base. According to this theory NH3 can be a base, even though it does not contain OH-.

4. Conjugate Acid-Base Pairs

- When an acid donates a proton, what remains is its conjugate base: the conjugate base of HCl is Cl-.
- When a base accepts a proton, what forms is its conjugate acid: the conjugate acid of NH3 is NH4+.
- The conjugate base of H2SO4 is HSO4- and the conjugate acid of H2O is H3O+.
- Water is amphoteric: it has both a conjugate acid H3O+ and a conjugate base OH-.

5. Lewis Concept

According to Lewis, a base is one that can donate a pair of electrons to form a coordinate covalent bond, and an acid is one that accepts a pair of electrons. Example: CN- is a Lewis base. NH3 is a base according to both the Bronsted-Lowry and Lewis theories.

6. Amphoteric Substances

Substances that react with both acids and bases are called amphoteric substances. Water is an important example of this.

Concept of pH and pOH

LESSON 6

1. Ionization of Water

Pure water ionizes only very slightly, so it is considered a weak electrolyte:

- H2O <=> H+ + OH-, or 2H2O <=> H3O+ + OH-
- Ionization constant: Kw = [H+] [OH-] = 1 x 10^-14 (at 25 C)
- In pure water [H+] = [OH-] = 1 x 10^-7 mol dm^-3, so its pH is 7 and it is neutral.

2. Definitions of pH and pOH

- pH is the measure of the hydrogen ion concentration: pH = -log [H+]
- pOH = -log [OH-]
- pH + pOH = 14

Condition	[H+] and [OH-]	pH
Acidic solution	[H+] > [OH-]	Less than 7
Neutral solution	[H+] = [OH-]	7
Basic solution	[OH-] > [H+]	More than 7

3. pH of Everyday Things

Thing	pH
Lemon juice	2.3
Soda water	3.8
Black coffee	5.0
Milk	6.3 to 6.6
Washing soda	12.0

4. Calculation Examples

- If a solution has [H+] = 1 x 10^-8 mol dm^-3, then pH = -log 1 x 10^-8 = 8, so the solution is basic.
- For SM NaOH, [OH-] = 5, pOH = -log 5 = -0.70, and pH = 14 - (-0.70) = 14.70.
- If a solution has pH 1.05, then pOH = 14 - 1.05 = 12.95 and [H+] = 10^-1.05 = 8.9 x 10^-2 mol dm^-3.

5. Indicators and Titration

- An indicator is a weak organic acid or base which changes its colour over a small range of pH.
- Titration is a technique in which a solution of known concentration is used to determine the concentration of an unknown solution.

6. Buffers

A buffer is an aqueous solution which resists a change in its pH on the addition of a small amount of acid or base. The composition of a buffer is a weak acid and its salt (its conjugate base), for example CH3COOH and CH3COONa, or a weak base and its salt, for example NH3 and NH4Cl. The serum of blood stays at about pH 7.4, which is due to buffers. Preservatives are also used to preserve food.

Salts

LESSON 7

1. What is a Salt?

A salt is an ionic compound in which the cation comes from the base and the anion comes from the acid. It is obtained along with water in the neutralization of an acid and a base. Examples: NaCl, K2SO4, KNO3, BaCl2 and KCl are salts, whereas HCl and HNO3 are acids and NaOH and KOH are bases.

2. Neutralization

The reaction of an acid and a base to form salt and water is called neutralization. Balancing it means that the H+ obtained from the acid and the OH- obtained from the base are in equal numbers:

- NaOH + HCl -> NaCl + H2O (one H+ and one OH-)
- 2NaOH + H2SO4 -> Na2SO4 + 2H2O (2NaOH to make two H+ and two OH- equal)
- Ca(OH)2 + 2HNO3 -> Ca(NO3)2 + 2H2O

3. Types of Salts

Type	pH	Examples
Acidic salts	Less than 7	NH4Cl, NaHSO4
Basic salts	More than 7	CH3COONa, Na2CO3
Neutral salts	7	NaCl, KCl

4. Preparation of Salts

- Action of acids on metals
- Action of acids on metal oxides, metal hydroxides, metal carbonates and metal bicarbonates
- Action of a base with an acid (neutralization)
- Action of a metal with a base

5. Uses of Salts in Daily Life

- As fertilizers in agriculture: ammonium chloride and NPK fertilizers
- As disinfectants: copper sulphate and potassium permanganate
- Plaster of Paris: to make plaster casts for broken bones
- Iron sulphate: in medicine for anaemia
- Sodium hydrogen carbonate: in medicine for stomach acidity
- Barium sulphate: in X-rays of the stomach
- Common salt: in food for taste and preservation

Organic Chemistry

UNIT 3

Organic Compounds

LESSON 8

1. What is Organic Chemistry?

The branch of chemistry which deals with hydrocarbons and their derivatives is called organic chemistry. Carbon is an essential element in organic compounds.

2. Vital Force Theory

According to the old idea, organic compounds can only be prepared in living tissues by a vital force and cannot be prepared in the laboratory. This is called the Vital Force Theory.

3. Catenation

Catenation is the ability of carbon atoms to join with one another through covalent bonds to form long chains or rings. Examples: in ethane two carbon atoms are joined to each other; in pentane there is a chain of five carbon atoms; in cyclohexane carbon atoms are joined in a ring. Carbon has a much greater power of catenation than silicon, because the C-C bond, at 355 kJ mol^-1, is much stronger than the Si-Si bond at 200 kJ mol^-1. Silicon is found in nature as silica and silicates, because the Si-O bond, at 452 kJ mol^-1, is stronger than the C-O bond at 351 kJ mol^-1.

4. Diversity and Magnitude of Organic Compounds

Organic compounds are very many and of very many kinds: more than 10 million organic compounds exist, far more than the compounds of all the other elements. The reasons are: catenation, isomerism, the strong covalent bonds of carbon, and the ability to form multiple bonds.

5. Important Characteristics of Organic Compounds

- Carbon is an essential element in them and they are derivatives of hydrocarbons.
- They dissolve in organic solvents.
- Their covalent bonds are weaker than ionic bonds, so they have low melting and boiling points.
- Their rate of reaction is slow, because their covalent bonds take time to break and need more energy; in contrast, ionic compounds react fast.
- On combustion, all organic compounds give carbon dioxide.
- Many organic compounds are used in perfumes, dyes and drugs.

6. Homologous Series

A homologous series is a group of compounds whose members have the same functional group, the same general formula, and differ from one another by CH2. Common series are: alkanes, alkenes, alkynes, alcohols and carboxylic acids.

Sources of Organic Compounds

LESSON 9

1. Main Sources

There are two main sources of organic compounds: living things and minerals.

- From living things: animals give proteins and fats, and plants give carbohydrates, proteins, vitamins and oils.
- From minerals: coal, petroleum and natural gas are obtained.

2. Coal

Coal is formed from trees buried in the earth. The percentage of carbon in it increases with its grade:

Type	Carbon
Wood	40 percent
Peat	60 percent
Lignite	70 percent
Anthracite	90 percent

3. Petroleum

Petroleum is a complex mixture of hydrocarbons formed underground from dead plants and animals. It is separated by fractional distillation into separate useful fractions, for example petrol, kerosene and diesel, all of which are organic compounds.

4. Natural Gas

Natural gas is mostly methane. In Pakistan it is called Sui gas.

Uses of Organic Compounds

LESSON 10

1. General Uses

Organic compounds are used everywhere in our daily life:

- As food
- As clothing
- As houses (shelter)
- As fuel
- As medicines
- As raw materials

2. Industrial Uses

- In making perfumes
- In making dyes
- In making soaps
- In making plastics — polyethene is a plastic
- In making drugs

3. Role of the Pharmaceutical Chemist

The pharmaceutical chemist plays an important role in laboratory analysis, quality assurance and the production of new effective drugs.

4. Compounds from Natural Sources

- Acetic acid is obtained from vinegar.
- Petrol, diesel and kerosene are used as fuel, all of which are organic compounds.

Alkanes and Alkyl Radicals

LESSON 11

1. Saturated and Unsaturated Hydrocarbons

Type	Bond	Example and General Formula
Saturated (alkanes)	Single bonds between carbon atoms	CnH2n+2, for example propane CH3CH2CH3
Unsaturated (alkenes)	Double bond C=C between carbon atoms	CnH2n
Unsaturated (alkynes)	Triple bond C#C between carbon atoms	CnH2n-2

2. Alkanes up to Decane

Name	Formula	Name	Formula
Methane	CH4	Hexane	C6H14
Ethane	C2H6	Heptane	C7H16
Propane	C3H8	Octane	C8H18
Butane	C4H10	Nonane	C9H20
Pentane	C5H12	Decane	C10H22

3. Types of Formulae

- Pentane: molecular formula C5H12, condensed formula CH3(CH2)3CH3, structural formula CH3-CH2-CH2-CH2-CH3.
- Octane: molecular formula C8H18, condensed formula CH3(CH2)6CH3, structural formula CH3-CH2-CH2-CH2-CH2-CH2-CH2-CH3.

4. Alkyl Radicals

An alkyl radical is formed by removing one hydrogen atom from an alkane. Its general formula is CnH2n+1.

- Methyl CH3- from methane CH4
- Ethyl CH3CH2- from ethane C2H6
- Propyl C3H7- from propane

Names of Simple Alkanes, Alkenes and Alkynes

LESSON 12

1. What is Nomenclature?

Nomenclature is the system of giving systematic names to compounds. By it, every compound has one accepted name.

2. Prefixes (Number of Carbons)

Carbons	Prefix	Carbons	Prefix
1	Meth	6	Hex
2	Eth	7	Hept
3	Prop	8	Oct
4	But	9	Non
5	Pent	10	Dec

3. Suffixes

- For alkanes -ane: methane, ethane, propane...
- For alkenes -ene: ethene, propene...
- For alkynes -yne: ethyne, propyne...

4. Identifying the Name from the Formula

- C2H4 — follows the pattern CnH2n, so it is an alkene: ethene
- C3H6 — propene
- C6H12 — hexene
- C8H16 — octene
- C3H4 — follows the pattern CnH2n-2, so it is an alkyne: propyne
- C5H8 — pentyne, C7H12 — heptyne, C6H10 — hexyne

5. IUPAC Rules for Alkynes

- Select the longest continuous chain containing the triple bond.
- Number the chain from the end nearest the triple bond.
- Name = prefix for the number of carbons + yne, for example ethyne, propyne.
- If there are branches, their names and position numbers are written first.

Introduction to Functional Groups

LESSON 13

1. What is a Functional Group?

A functional group is an atom or group of atoms that gives a molecule its characteristic properties. Compounds are divided into separate families on the basis of their functional group.

2. Functional Groups Containing Carbon, Hydrogen and Oxygen

Functional Group	Formula	Example
Alcohol	-OH	CH3CH2OH
Ether	-O-	CH3OCH3
Aldehyde	-CHO	CH3CHO
Ketone	-CO-	CH3-CO-CH3
Carboxylic acid	-COOH	CH3-COOH
Ester	-COOR	—

3. Identification Examples

- In CH3-CHO there is -CHO, the aldehyde group.
- In CH3-CH2-CH2-OH there is -OH, the alcohol group.
- In CH3-CO-CH3 there is -CO-, the ketone group.
- In CH3-COOH there is -COOH, the carboxylic acid group.
- In CH2=CH-CH3 there is C=C, the alkene double bond.

4. Functional Group and Property

If blue litmus paper dipped in a solution turns red, the solution is acidic, so it has the carboxylic acid group -COOH, which gives the acidic character. Remember that -COOH is used for carboxylic acids.

Biochemistry

UNIT 4

Carbohydrates

LESSON 14

1. What are carbohydrates?

Carbohydrates are organic compounds containing carbon, hydrogen and oxygen, with the general formula $C_x(H_2O)_y$. They contain an aldehyde group -CHO or a ketone group >C=O along with many hydroxyl groups -OH. Examples: glucose, fructose, starch, cellulose.

2. Types of carbohydrates

Type	Definition	Example
Monosaccharides	Simple sugars, cannot be simplified further on hydrolysis	Glucose, fructose
Oligosaccharides	Give 2 to 10 monosaccharides on hydrolysis	Sucrose, lactose
Polysaccharides	Give more than 10 monosaccharides on hydrolysis	Starch, cellulose, glycogen

- Glucose is an aldose because it contains an aldehyde group.
- Lactose is called milk sugar.
- Starch gives glucose on hydrolysis.
- Cotton contains about 95 percent cellulose.

3. Sources

- Monosaccharides: in grapes, honey and guava
- Oligosaccharides: in sugarcane, sugar beet, milk, wheat and vegetables
- Polysaccharides: in the cell walls of all land plants, in cotton, in cereal foods and in potato

4. Uses

- They are a source of energy for survival.
- They sustain the structure of plants.
- They are a storage of energy in the form of starch and glucose.
- They keep blood sugar levels in check.
- Sucrose is used as a food additive.
- Cellulose gives food its bulk and fibre and promotes peristalsis in the intestine.
- Cellulose is a raw material in textiles and paper; starch is used in the production of ethanol and in laundries.

Proteins

LESSON 15

1. What are proteins?

Proteins are polymers of amino acids, joined to one another by peptide bonds. One protein may contain from 60 to 6000 amino acid molecules. A protein is a hetero polymer and a polyamide.

2. Amino acids

Amino acids are the building blocks of proteins. They have an amino group -NH_2 and a carboxyl group -COOH on the same carbon atom. General structure: a central carbon atom joined to H, -NH_2, -COOH and a side chain R.

3. Peptide bond

A peptide bond is the covalent bond that joins two amino acids to each other, between the carboxyl group of one and the amino group of the other.

Type	Number of amino acids
Dipeptide	2
Tripeptide	3
Polypeptide	Many

4. Functions of proteins

- Proteins are the building blocks of the body.
- As enzymes, they catalyze biological reactions.
- As haemoglobin, they carry oxygen.

Lipids

LESSON 16

1. What are lipids?

Lipids include fats and oils. Both are esters of fatty acids and glycerol.

2. Difference between fats and oils

Fats	Oils
Are solids at room temperature	Are liquids at room temperature
Increase the cholesterol level in the body	Are generally obtained from plants
Examples: butter, ghee	Examples: sunflower oil, olive oil

3. Sources

- From animals: marine animals like salmon and whales, butter, ghee and cheese
- From plants: sunflower, coconut, ground nuts, corn, cotton seed and olive

4. Uses

- Lipids act as transporters of fatty acids in the body.
- They are transporters of fat-soluble vitamins A, D, E and K.
- Some lipids reduce the cholesterol level.
- Fats and oils are used for cooking and frying food.
- They are used in detergents, soaps, cosmetics, polishes and paints.
- They activate the enzymes.
- Butter and ghee are used in bakery products and sweets.

Nucleic Acids

LESSON 17

1. What are nucleic acids?

Nucleic acids are macromolecules formed by the polymerization of nucleotides. Each nucleotide is composed of a pentose sugar, a phosphate group and a nitrogenous base. They store and transmit hereditary information.

2. DNA

- DNA (deoxyribonucleic acid) is the store house of information.
- It stores and transmits hereditary information from parents to children.
- It encodes genetic information and transfers it from generation to generation.
- Present in the nuclei of cells, it carries the information for making organs like heart, liver, eyes, kidneys, legs and head.
- DNA is the most important thing in any living organism.

3. RNA

- RNA (ribonucleic acid) receives the information obtained from DNA.
- It decodes, reads and uses the information.

DNA	RNA
It is the store house of information	It puts the information to use
It encodes genetic information and transfers it from generation to generation	It receives, decodes and reads the information

Vitamins

LESSON 18

1. What are vitamins?

Vitamins are accessory growth factors which are needed in small amounts for the growth and health of the body.

2. Types of vitamins

Type	Vitamins	Property
Fat-soluble	A, D, E, K	Stored in the body; excess can cause harm
Water-soluble	B complex, C	Not stored in the body; excess does not cause harm

3. Role of each vitamin

- Vitamin A: necessary for sight
- Vitamin D: for bones; helps in the use of calcium and phosphorus
- Vitamin E: for protection of cells
- Vitamin K: for clotting of blood
- Vitamin B complex: for energy and nerves
- Vitamin C: for skin and immunity

4. Vitamin D in particular

Vitamin D is a fat-soluble vitamin. It is necessary to keep bones and teeth strong. Its deficiency causes rickets in children and weakness of bones in adults. Excess of vitamin D in the body causes calcification and pain in bones.

5. Deficiency diseases and harms of excess

Vitamin	Deficiency	Excess
A	Weakness of sight	Irritation and headache
D	Rickets	Calcification and pain in bones
B complex	Nerve and energy problems	—
C	Scurvy	—
E	—	Fatigue and headache
K	Delay in clotting of blood	Liver and kidney diseases

Environmental Chemistry-I: Atmosphere

UNIT 5

Composition of the Atmosphere

LESSON 19

1. What is the Atmosphere?

The atmosphere is the blanket of gases around the earth that covers the earth and is held to the earth by gravity. It is necessary for life and controls weather, air and temperature.

2. Composition of the Atmosphere

Gas	Percentage
Nitrogen	78.09 percent
Oxygen	20.95 percent
Argon	0.93 percent
Carbon dioxide	0.03 percent

Besides these there are water vapour, ozone, helium, methane and other small gases. Around 80 percent of the mass of the atmosphere is in the troposphere.

Layers of the Atmosphere

LESSON 20

1. Layers of the Atmosphere

There are 5 main layers of the atmosphere, from the earth upward in this order: troposphere, stratosphere, mesosphere, thermosphere and exosphere.

2. Troposphere

- It is the lowest layer, from the earth's surface to about 11 to 12 kilometres.
- It contains about 75 to 80 percent of the mass of the atmosphere.
- Weather occurs in this layer and almost all aeroplanes pass through this layer.
- The temperature falls with height, from 15 C to -56 C.
- There is much air movement and an active weather system in it.

3. Stratosphere

- It is the second layer, from 12 to about 50 to 55 kilometres.
- It contains the ozone layer, which protects from ultraviolet radiation.
- The temperature rises with height, because ozone absorbs radiation from the sun and gives heat.
- Air movement is low and aeroplanes do not go in this layer.

4. Other Layers and Boundaries

- The boundary separating the troposphere and the stratosphere is called the tropo-pause.
- The mesosphere is above the stratosphere.
- The thermosphere is the second highest layer of the earth's atmosphere; the highest is the exosphere.
- Ozone gas is found in the stratosphere.

Feature	Troposphere	Stratosphere
Height	0 to about 12 km	12 to 50-55 km
Temperature with height	Falls (15 to -56 C)	Rises
Special feature	Weather and aeroplanes	Ozone layer
Air movement	Much	Low

Pollutants

LESSON 21

1. Primary and Secondary Pollutants

- Primary pollutants come directly from sources, for example CO2 from vehicles and SO2 from factories.
- Secondary pollutants are formed in the atmosphere by chemical reactions of primary pollutants, for example acid rain and ozone.

2. Major Air Pollutants and Their Sources

Pollutant	Source
Carbon dioxide, sulphur dioxide, oxides of nitrogen	Vehicles and factories
Smoke and dust	Burning of fuel
Pesticides and fertilizer particles	Land
Ozone and acid rain	Secondary reactions

3. Effects on Human Health

- Air pollutants affect the respiratory systems of humans and animals.
- Acid rain and smoke damage the eyes and lungs.
- Breathing polluted air can cause cough, asthma, lung diseases and allergy.
- CO2 and other greenhouse gases cause global warming.

Acid Rain and Its Effects

LESSON 22

1. What is Acid Rain and How is it Formed?

The cause of acid rain is sulphur dioxide SO2 and oxides of nitrogen NO2 from factories and vehicles. These gases combine with water vapour in the atmosphere to form acidic compounds and come down with the rain. It is a secondary pollutant problem.

2. Effects of Acid Rain

- It affects the respiratory system.
- It harms the aquatic ecology; the living organisms of lakes and rivers are affected.
- It corrodes water pipelines, so heavy metals like iron, lead and copper seep into drinking water.
- It damages stone and metal buildings and monuments.

Ozone Depletion and Its Effects

LESSON 23

1. Ozone Layer

Ozone O3 is a gas which is found in the stratosphere. The ozone layer absorbs the harmful ultraviolet radiation of the sun and protects life on earth.

2. Formation of Ozone

In the stratosphere, ozone is formed from oxygen by the action of the sun's radiation and at the same time it also keeps breaking down; this natural balance maintains the ozone layer.

3. Ozone Depletion

- Man-made chemicals, especially the gases used in refrigerators and air conditioners, break down ozone.
- As ozone decreases, more ultraviolet radiation reaches the earth.

4. Effects

- Ultraviolet radiation damages the eyes and lungs.
- It is harmful for the skin and other living organisms.
- Therefore protection of the ozone layer is necessary for the protection of life.

Greenhouse Effect

LESSON 24

1. What is the Greenhouse Effect?

Greenhouse gases trap the sun's heat in the atmosphere, due to which the earth stays warm. This natural process is necessary for life, but with an increase in the gases it becomes greater.

2. Greenhouse Gases

- Carbon dioxide CO2
- Methane CH4
- Nitrous oxide
- Water vapour

Note: Oxygen is not a greenhouse gas.

3. Global Warming

- With an increase in greenhouse gases, more heat is trapped, due to which the earth's temperature rises; this is called global warming.
- Global warming is the continuous rise in the average temperature of the earth.
- Results: the weather changes, ice melts, sea level rises and living organisms are badly affected.
- The rise in CO2 concentration is a major cause of global warming, which increases by the burning of fuel.

Environmental Chemistry-II: Water

UNIT 6

Water

LESSON 25

1. Presence of Water

- About 70 percent of the Earth's surface is covered by water.
- The percentage of fresh water is about 3 percent; most of the rest is the saline water of the oceans.
- Our dependence on water is very great and it is very important to maintain its quality.

2. Composition of Water

A water molecule is made up of one atom of oxygen and two atoms of hydrogen, connected by covalent bonds: H2O. Because of the higher electronegativity of oxygen, the oxygen carries a partial negative charge and the hydrogens carry partial positive charges, so water is a polar molecule. Hydrogen bonding exists between water molecules.

3. Properties of Water

- Water is colourless, odourless and tasteless.
- Water is a polar solvent.

4. Why is Water a Universal Solvent?

Water is called the universal solvent, because it dissolves very many substances. The reason is that it is polar: it dissolves ionic compounds like NaCl and many molecular substances. The hydrogen bonding between water molecules also helps its solvent character. Water is important in both the environment and industry.

Soft and Hard Water

LESSON 26

1. Soft and Hard Water

- Soft water is water that forms lather easily with soap and contains very few salts of calcium and magnesium.
- Hard water is water that does not form lather easily with soap, because it contains dissolved salts of calcium and magnesium — chlorides, sulphates and bicarbonates.

2. Types of Hardness

Type	Cause	Removed by Boiling?
Temporary hardness	Ca(HCO3)2 and Mg(HCO3)2	Yes
Permanent hardness	Chlorides and sulphates of calcium and magnesium, such as MgSO4	No

3. Methods to Remove Temporary Hardness

- Boiling: on heating, the bicarbonate breaks down into insoluble carbonate which settles at the bottom.
- Clark's method: adding slaked lime Ca(OH)2 converts the bicarbonate into carbonate which settles at the bottom.
- Adding washing soda Na2CO3.

4. Methods to Remove Permanent Hardness

- Adding washing soda Na2CO3: calcium and magnesium settle at the bottom as insoluble carbonates.
- The sodium zeolite method: water is passed through zeolite which exchanges calcium and magnesium for sodium.
- The ion exchange resin method.

5. Disadvantages of Hard Water

- It does not form lather with soap, so soap is wasted and clothes are not cleaned.
- It forms scale in boilers and pipes, which wastes energy and damages machines.
- It affects the taste and cooking of food.
- It can harm the skin and hair.

Pollutants of Water

LESSON 27

1. Definition of a Water Pollutant

A water pollutant is a substance that mixes in water and makes it unfit for drinking and other uses, for example chemical waste, sewage, pesticides, fertilizers and germs.

2. Industrial Pollutants

Industrial pollutants are the waste from factories:

- Chemicals
- Heavy metals like lead, mercury and cadmium
- Oil, dyes and hot water

They mix into rivers and land, kill aquatic organisms, poison drinking water and pollute the land.

3. Domestic and Agricultural Pollutants

- Domestic sewage, which contains germs
- Pesticides and fertilizers which wash off the land and mix into water
- They spoil the quality of water and spread diseases

4. Effects

- Aquatic organisms die
- Drinking water becomes poisonous
- The land is polluted
- Water-borne diseases spread

Water-Borne Diseases

LESSON 28

1. Names of the Diseases

The water-borne diseases are: diarrhoea, dysentery, cholera, cryptosporidium, fluorosis, hepatitis, hookworm, jaundice and typhoid.

2. Types by Cause

Cause	Diseases
Bacteria	Cholera, typhoid
Virus	Hepatitis, polio; diarrhoea and dysentery are also caused by bacteria or viruses
Worms	Hookworm
Chemical (excess amount of fluoride)	Fluorosis

3. How Do the Diseases Occur?

These diseases are caused by drinking or using polluted water. When sewage mixes into drinking water, germs enter the body.

4. Prevention

- Drink clean water.
- Boil or filter water.
- Keep sewage away from drinking water.

Analytical Chemistry

UNIT 7

Qualitative and Quantitative Analysis

LESSON 29

1. What is Analytical Chemistry?

Analytical chemistry is the branch of chemistry that deals with the instruments and methods to separate, identify and quantify matter. It is used in every type of chemistry: in organic, inorganic, physical and biochemistry, analytical methods are needed to identify samples and determine their quantity. Therefore it is a part of all types of chemistry.

2. Two Types

Qualitative Analysis	Quantitative Analysis
Tells which substances are present in a sample	Tells in what quantity those substances are present
Example: chemical and flame tests	Example: titrimetric, gravimetric and combustion analysis

3. First Step: Identification

The first step to quantify any sample is to identify it. Because first it is necessary to know which compounds are in the sample, and only then can their quantity be determined. If the compound is not identified, the method to measure its quantity cannot be chosen. Therefore first identification, then quantity.

4. Types of Qualitative Analysis

- Organic qualitative analysis: deals with the identification of the presence of functional groups in compounds.
- Inorganic qualitative analysis: identification of ions and elements, such as the flame test. The flame test of copper halide with bluish-green colour identifies the presence of halogen.
- In qualitative analysis the sample may be solid, liquid, gas or a mixture.

Important Parameters

LESSON 30

1. Error

- **Systematic (determinate) error:** an error that occurs in the same way every time and can be estimated and eliminated. Example: improper functioning of an instrument, and improper functioning in acid-base titration, such as a wrong burette reading or a wrong indicator — these produce an error in the same direction in every measurement.
- **Indeterminate error:** an error that occurs randomly and is different every time.
- Examples of errors in everyday life: a wrong weight on a weighing machine, a clock running fast or slow, measuring a wrong length with a measuring tape, reading a wrong temperature on a thermometer.

2. Accuracy and Precision

Term	Meaning
Accuracy	Agreement between a measured value and the accepted true value
Precision	Agreement between replicate measurements of the same quantity, that is the repeatability of a result

A measurement can be precise but not accurate, accurate but not precise, both or neither.

3. Example: Volumes of a Gas (True Value 32 cm^3)

Set	Values	Result
First	32, 45, 17, 23	Neither — the values are scattered
Second	32, 33, 34, 35	Accurate but not precise — close to the true value but not repeatable
Third	32, 45, 45, 32	Neither — the values are scattered
Fourth	32, 32, 33, 32	Accurate and precise both — repeatable and close to the true value

Classical Methods

LESSON 31

1. What are Classical Methods?

Classical methods are traditional analytical methods based on chemical reactions and simple measurements. They are simple and cheap but slow and need a large amount of sample.

2. Titration

Titration is the technique in which the concentration of an unknown solution is determined by comparing the volume of a solution of known concentration with the volume of the solution of unknown concentration.

- **Burette:** which contains the titrant (solution of known concentration).
- **Flask:** which contains the analyte (substance being analysed).
- **Pipette:** for taking a measured quantity.
- **Indicator:** which changes its colour at the end point and tells that the titration is complete and the equivalence point has been reached.

3. Titrimetric and Gravimetric Analysis

Titrimetric	Gravimetric
Based on the measurement of volume	Based on the measurement of weight
The volume of a solution of known concentration is used	The substance is converted into a product and then weighed

4. Steps of Gravimetric Analysis

- Preparation of a solution with known weight of sample
- Separation of the desired constituent
- Weighing the separated constituent
- Computation of the amount of the separated constituent in the sample

5. Comparison of Classical and Instrumental Methods

Techniques are compared on the basis of speed, accuracy, precision, amount of sample, cost and training. The classical method is simple and cheap but slow; the instrumental method is fast, sensitive and expensive and needs specialized training.

Modern Instrumental Methods

LESSON 32

1. Instrumental Methods

Instrumental methods measure the physical properties of substances with modern instruments. They are more effective than classical methods because they are very fast, sensitive, work with a small amount of sample, and give both identification and quantity in less time. They can also analyse trace amounts and work automatically. In contrast, classical methods are slow and accuracy decreases with decreasing amount of sample.

2. Spectroscopy

- Spectroscopy is the interaction of light with matter.
- It is used to assess the concentration or amount of a given atomic, molecular or ionic chemical.
- Physical methods deal with the physical properties of substances, such as atomic emission spectroscopy, Fourier transform spectroscopy and X-ray spectroscopy.
- Ultraviolet and visible spectroscopy and infrared spectroscopy are its types.
- Infrared spectroscopy tells the identification and quantity of samples in less time and at less cost, and no polluting chemical is required for it; it is used for the identification of functional groups in food products, polymers and industries and for quality control.

3. Chromatography

- Chromatography is a method for the separation of compounds; types: high performance liquid chromatography (HPLC) and gas chromatography.
- Gas chromatography is a method for the separation of gases and volatile compounds; the separation takes place by the exchange between a mobile gas phase and a liquid or solid stationary phase. Gas is the mobile phase in it.
- Instruments: gas cylinder, sample injector, gas chromatograph column, detector and data collection device. The detector detects the elution of compounds and sends signals to the computer.
- Uses: inorganic compounds, carbohydrates, proteins, lipids, vitamins, pollutants like benzene, plastic materials and dairy products.

4. Electrochemical Methods

- These methods measure the potential, charge and electrical quantity of a solution, and these measurements are carried out with an electrochemical cell.
- The cell has two half cells with anode and cathode electrodes, connected by a salt bridge. At the anode electrons are liberated in oxidation and at the cathode electrons are consumed in reduction.
- **Potentiometry:** the potential between two electrodes is measured by a voltmeter and there is no flow of current; it is used to find the concentration of a solute in solution, pollutants in water, pharmaceuticals and drugs, quality control in the food industry and clinical chemistry.
- **Applications of conductometry:** the dissociation constant can be determined, the solubility of a sparingly soluble salt can be determined, the rate constant of a reaction can be studied, and the end point of titration can be determined.

Industrial Chemistry

UNIT 8

Preparation of Soap

LESSON 33

1. What is Soap?

Soaps are salts of fatty acids. Soap molecules reduce the surface tension of water; such substances are called surfactants.

2. Two Ends of the Soap Molecule

End	Property
Carboxylate end	Hydrophilic — is attracted to water
Hydrocarbon end	Hydrophobic — is attracted to oil and dirt

3. Saponification

Saponification is the chemical reaction in which fat or oil reacts with caustic soda NaOH or caustic potash KOH to form soap and glycerine.

- NaOH gives hard soap.
- KOH gives soft soap, because potassium soaps are more soluble in water.

4. Stages of Preparation (Flow Sheet)

- 1. Oil or fat
- 2. Bleaching, refining, deodorization
- 3. Saponification — aqueous NaOH is added
- 4. Separation — NaCl brine is added and glycerine is separated
- 5. Purification — citric acid is added
- 6. Finishing — fragrances, conservative agent and colouring are added
- 7. Soap is prepared

Preparation of Sugar from Sugarcane

LESSON 34

1. Raw Material

The raw material for the preparation of sugar is: sugarcane, lime and water. Harvesting of sugarcane is the most important step in the preparation of sugar.

2. Stages of Preparation

- 1. Harvesting and delivery: Sugarcane is harvested in the cooler months and sent to the factory.
- 2. Juice extraction: The cane is cut into chips and the juice is extracted in mills.
- 3. Clarification: Heat, lime and flocculation aids are added to the juice; lime is a suspension of calcium hydroxide which kills enzymes and brings the pH from 5.0 to 6.5 to neutral; mud is separated. Bagasse is the fibre which does not dissolve in water.
- 4. Concentration: The clarified juice is heated in evaporators and sucrose is increased from 10 to 15 percent to 55 to 59 percent.
- 5. Crystallization: The syrup is further evaporated in vacuum pans, seed crystals are added, and sugar crystals form.
- 6. Crystal separation and drying: The crystals are separated from molasses in a basket-type centrifuge machine and dried.

Preparation of Soft Drinks

LESSON 35

1. Components of Soft Drinks

The basis of soft drinks is syrup, which is made up of water, sugar, acid, colouring and flavouring agents. The components are:

- Water, calcium and other minerals
- Colouring and flavouring agent
- Sugar for microbial growth
- Citric acid for sour taste

The syrup is prepared by dissolving the ingredients into water to 65 Brix.

2. Stages of Preparation

- Stage 1 — Water cleaning: Raw water is passed through a filter, water softener, chemicals and pump.
- Stage 2 — Heating, evaporation and carbonation: The water is heated with electric heaters, passed through a cooling pipe, and carbon dioxide is added to make carbonated water.
- Stage 3 — Mixing: Colouring, syrup and flavour are added in the mixing tank.
- Stage 4 — Filtering and filling: The drink is filtered and filled into bottles and cans.
- Stage 5 — Packaging: Packed in cartons and sent to the supermarket.

Petroleum Industry

LESSON 36

1. What is Petroleum?

Petroleum is a natural substance trapped in rocks beneath the earth's crust. Petroleum means rock oil. It is a complex combination of gaseous, liquid and solid hydrocarbons, in which water, salts and earth particles are also present. It is a liquid lighter than water and does not dissolve in it.

2. Black Gold

Petroleum is called Black Gold because it is black and as valuable as gold. From it petrol, diesel, kerosene, gas, plastics and many other things are obtained. It is the main source of the economy and energy of countries.

3. Fractional Distillation and Fractions

Petroleum is separated into fractions by fractional distillation, which is done in refineries in fractionating columns — the column is cold at the top and hot at the bottom.

Fraction	Boiling range (C)	Carbons	Use
Refinery gas	-160 to -5	1 to 4	Home heating and cooking
Petrol	40 to 110	5 to 8	Car fuel
Naphtha	110 to 180	8 to 10	For plastics
Kerosene	180 to 260	10 to 16	Jet aircraft fuel
Diesel	260 to 320	16 to 20	Fuel for buses and lorries
Fuel oil	320 to 400	20 to 50	Industrial heating
Bitumen	400 to 600	More than 50	Surfacing roads

Pharmaceutical Industry

LESSON 37

1. What is the Pharmaceutical Industry?

The pharmaceutical industry researches, manufactures, sells and distributes drugs for the prevention, treatment and cure of illnesses.

2. Importance

- Treatments increase life expectancy.
- The objective to eradicate diseases is achieved, for example the elimination of smallpox.
- Pain and suffering are reduced.
- Vaccines save money, because prevention is cheaper than treatment.
- Hospital stays are shorter.
- The industry employs millions of people.
- Pharmaceutical companies boost the global economy.